STAFF REVIEW · H446 030 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. The Boolean expression becomes simpler. Variables that change within the group can be removed, leaving only variables that stay the same.

2. Step 1: 13 in binary is 1101. Step 2: pad the magnitude to seven bits: 0001101. Step 3: because the number is negative, put 1 in the sign bit. Answer: 1 | 0001101 = 10001101. This is sign and magnitude , not two’s complement. SIGN · 1 BIT 1 Negative MAGNITUDE · 7 BITS 0001101 13 in binary

3. Insert the values in the order given . 40 becomes the root. 20 goes left of 40; 60 goes right of 40. 10 goes left of 20; 30 goes right of 20; 50 goes left of 60. Every smaller value goes to the left and every larger value to the right. 40 20 60 10 30 50

4. Align exponents: B = $0.0100_2 \times 2^3$. Subtract mantissas: $0.1100_2 - 0.0100_2 = 0.1000_2$. Answer: $0.1000_2 \times 2^3 = 4$.

I Do — source worked steps

1. Rightmost result bit — CODE:
Sum = 0

2. Next column — CODE:
Carry = 1

3. Trace XOR and AND — CODE:
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry

4. Connect both half adders — CODE:
S_1 = A XOR B
Sum = S_1 XOR Cin
Cout = (A AND B) OR (S_1 AND Cin)

We Do — guided CHECK

1. Complete the same formulae
CODE:
S = A XOR B
C = A AND B

2. Fill the same table
CODE:
A B | S C
0 0 | ? ?
0 1 | ? ?
1 0 | ? ?
1 1 | ? ?

CODE:
A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1

3. Improve the same explanation
CODE:
Two half adders plus an ? gate can form a full adder.
CODE:
Two half adders plus an OR gate can form a full adder.

4. Complete the output
CODE:
1 + 1 + 1 = binary ?
Sum = ?
CarryOut = ?
CODE:
1 + 1 + 1 = binary 11
Sum = 1
CarryOut = 1

You Do — source mark schemes

1. Half-adder question
CODE:
S = A XOR B
C = A AND B

A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1

Two half adders can be joined, with an OR gate combining carry outputs, to form a full adder.
Check: Use XOR for Sum, AND for Carry and mention the OR connection.

2. Complete eight cases
CODE:
A B Cin | S Cout
0 0 0 | 0 0
0 0 1 | 1 0
0 1 0 | 1 0
0 1 1 | 0 1
1 0 0 | 1 0
1 0 1 | 0 1
1 1 0 | 0 1

1 1 1 | 1 1
Check: The outputs match ordinary binary addition of three bits.

3. Explain four-bit addition
CODE:
Use one full adder for each bit position. The carry-out from a lower-order full adder is connected to the carry-in of the next higher-order full adder. This propagates carries across the four-bit addition.
Check: Your answer connects carry-out from one bit to carry-in of the next.

4. Add 0111 + 0001
CODE:
0111
+ 0001

1000

A carry is generated at the least-significant column and propagates through the next two columns.
Check: The final result is 1000.

Exit Ticket — source answer

1. True. A half adder uses XOR for Sum and AND for Carry.